

WEEK 6 — TASK 5 REPORT

Model Evaluation: Customer Churn Prediction

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Random Forest Classifier

Evaluation of the final machine learning model using classification metrics, prediction analysis and business interpretation.

1. Objective

The objective of Task 5 is to evaluate the trained Customer Churn Prediction model using appropriate classification metrics and interpret its performance. The evaluation focuses especially on the Churn class because correctly identifying customers who may leave is important for customer retention.

2. Test Data

The trained Random Forest Classifier was evaluated on **1,409 unseen test customers**. The 80:20 train/test split was stratified so that the relative proportion of churn and non-churn customers was preserved between training and testing data.

3. Evaluation Metrics

Accuracy measures the proportion of all predictions that were correct. **Precision** for Churn measures how many customers predicted to churn actually churned. **Recall** for Churn measures how many of the customers who actually churned were successfully identified. **F1-score** combines precision and recall into a single measure. The confusion matrix provides the counts of correct and incorrect predictions for each class.

Metric	Result
Accuracy	79%
Precision — Churn	64%
Recall — Churn	50%
F1-score — Churn	56%
Precision — No Churn	83%
Recall — No Churn	90%
F1-score — No Churn	86%
Macro-average F1	71%
Weighted-average F1	78%

4. Classification Report Interpretation

The model performed substantially better on the No Churn class than on the Churn class. No Churn recall was 90%, indicating that most customers who remained were correctly identified. Churn precision was 64%, while Churn recall was 50%, showing that the model's ability to detect actual churners is the main area requiring improvement.

5. Incorrect Predictions

The model made **294 incorrect predictions** out of 1,409 test samples. These errors can be divided into false positives and false negatives. False positives are customers predicted to churn who did not churn. False negatives are customers who actually churned but were predicted as No Churn.

False negatives are especially important in a churn-prediction setting because a missed churner may leave without receiving a retention intervention. This makes recall a particularly valuable metric for future model optimization.

6. Confusion Matrix

A confusion matrix was generated in the notebook with Actual classes on the rows and Predicted classes on the columns. It provides a direct view of true negatives, false positives, false negatives and true positives. The matrix should be included with the final submission screenshot as **Task5_Confusion_Matrix.png**.

7. Model Performance Analysis

The overall accuracy of 79% is a reasonable baseline for the project, but accuracy alone does not fully describe the model because the dataset contains more No Churn customers than Churn customers. The 50% recall for Churn demonstrates that a substantial number of actual churners are still being missed.

The 64% precision for Churn indicates that churn predictions are useful but not perfectly reliable. In a real organization, the predictions should therefore be treated as a prioritization tool rather than an automatic decision system.

8. Business Interpretation

A telecom company could use the model to identify customers who may be at risk of leaving and prioritize them for retention campaigns. Customers predicted as high risk could receive personalized offers, contract incentives, service assistance or targeted customer-support outreach.

Because missing actual churners is costly, future model development should emphasize increasing recall while maintaining acceptable precision. Threshold tuning and class-weighted learning are two practical approaches.

9. Limitations

- The evaluation uses one train/test split.
- The Random Forest baseline was not extensively optimized.
- Class imbalance affects interpretation of overall accuracy.
- The model does not establish causal reasons for customer churn.
- Real-world deployment would require monitoring, calibration, privacy controls and periodic retraining.

10. Future Improvements

- Tune the classification threshold to improve churn recall.
- Use `class_weight` options or resampling methods.
- Perform cross-validation and hyperparameter search.
- Compare additional algorithms such as Logistic Regression, Gradient Boosting and XGBoost.
- Evaluate ROC-AUC and Precision-Recall AUC.
- Deploy a dashboard that displays customer churn risk and supporting factors.

11. Conclusion

Task 5 successfully evaluated the Customer Churn Prediction model. On 1,409 unseen customers, the Random Forest Classifier achieved 79% accuracy, 64% churn precision, 50% churn recall and 56% churn F1-score. A total of 294 predictions were incorrect.

The model provides a useful baseline for churn prediction, but improving recall for the Churn class should be the main focus of future work.

References: Scikit-learn documentation; IBM Telco Customer Churn dataset; Pandas documentation; Matplotlib and Seaborn documentation.